GE 286 PRINCIPLES OF SURVEYING

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DEFINITION OF LAND SURVEYING

- The **science**, **art** and **technology** of determining the **relative positions** of **natural, man made** features above, on or beneath the **earth's surface** and the **representation** of these information either **graphically** or **numerically**.

- Majority of professionals think it is a process of measuring distances, angles and height differences on site for the preparation of large scale plans in order that engineering works are located at their correct positions on the ground.

- In a more general sense, surveying can be regarded as that discipline that includes all methods for

i. Measuring

ii. Processing and

iii. Dissemination of information about the physical earth and environment.

- The practice of surveying therefore involves:

i. Determination of the shape of the earth (the size, shape, position)
ii. Provision of plans, maps and charts covering the earth surface
iii. Positioning of objects in space,
iv. Positioning of physical features, structures and engineering works on, above or below the surface of the earth;
v. Determination of positions of boundaries of public or private land, including national and

and international boundaries, and the registration of these lands with appropriate authorities;

- Design, establish and administration of land and Geographic Information Systems- (collection, storage, analysis and manipulation of that data to produce maps, charts and reports for use in planning and design processes)

Basic surveying measurements and deliverables

1. Field operations involves: measuring distances, height differences and angles using ground-base or space base instruments and techniques.

The measured quantities are processed to determine:

i. horizontal positions of arbitrary points on the earth's surface;

ii. elevations or heights of any point above or below a reference surface, such as the MSL;

iii. Configuration of the ground;
iv. Lengths and directions of lines;
v. Areas of tracts bounded by given lines.

2. Setting-out design drawings from paper onto the ground to locate construction lines for buildings, bridges, highways, canals etc.

Geodetic and Plane Surveys

- With respect to assumptions and accuracies required, surveying may be divided into Geodetic and Plane.
- Geodetic- curved surface of the earth is considered by performing computations on an ellipsoid.
- Heights are referenced to the surface of the ellipsoid (ellipsoidal or geodetic heights)

- Plane surveying- relatively small areas of the earth are involved and the surface is considered to be horizontal. Angles measured are plane angles
- All height measurements are referenced to the mean sea level or the geoid and are termed Orthometric heights.

Classes of survey

- Topographic
- Cadastral
- Hydrographic
- Photogrammetric
- Mining
- Engineering surveying.

Topographical and Cadastral Surveys

- These are surveys conducted to determine the configuration of the ground as well as the location of the natural and man-made features of the earth including hills, valleys, railways etc.
- These are surveys conducted for legal purposes such as deed plans showing and defining legal property boundaries and the calculation of area(s) involved.

Hydrographic and Photogrammetric surveys

- These are surveys conducted to determine the position of the survey vessel, depth of water and to investigate the nature of the sea bed.
- **Photogrammetric Surveying**: Process of making precise measurement and creating detailed maps from aerial images or photographs.

Mining and Engineering surveys

- **Mining Surveys:** These are surveys executed to establish location and boundaries of mining claims. It also involves the establishment of underground workings horizontally, vertically and lay out shaft connections.

- **Engineering Surveying:** Surveys executed to locate or lay out engineering or building works such as roads, railways, tunnels, dams etc.

Plan Scales

Definition

- *A map/plan scale is the ratio between a distance on a map/plan and the corresponding distance on the terrain.*

Scale = Map distance/ Ground distance

- It is also an expression of how much the area represented on the ground has been reduced on the map.
- Scale is the single most critical mathematical feature of any map. Large, medium and small scale

Types of map scales

- **Verbal scale:** relationship between a map distance and a ground distance. E.g.*One cm represents 16 km.*
- **Graphic scale***,* or **Bar scale**: This shows directly on the map the corresponding ground distance. For example:

Another great thing about bar scales is that *they remain correct if the map is reduced or enlarged* photographically. This is not true of the other two types of scales.

- The main drawback of bar scales is that they are usually short compared to the map itself, and hence measuring longer distances is difficult.

- **Representative fraction**, or ratio scale. Compared to the first two, it is the most abstract, but also the most versatile. A representative fraction, or RF, shows the relationship between one of any unit on the map and one of the same unit on the ground. example 1/24,000, but are usually written with a colon, as in 1:24,000.

Conversion of scales to representative fractions

Example 1

- If 1 centimetre on a map represents 10 metres on the ground, what will be the RF?

  = 1:1000

- Ground units = 10 * 100 centimetres =1000 cm

- **Example 2**

If the RF of a map is I: 120 how many units on the ground do 2 map units represent?

- ***Solution***

RF = 1: 120

i.e., 1 unit = 120 units

Therefore 2 units = 2 x 120 unit

= 240 units

Conversion of areas by representative fractions

- **Ground Area = Plan Area x (Scale factor)^2**

Therefore Plan Area = $\frac{\text{Ground Area } 4}{4^2}$

- Therefore **Plan Area = Ground Area x (RF)^2**

- An area was measured on a plan as 250 mm x 175 mm. Calculate the ground area in metres if the scale is (a) 1:2000 (b) 1: 500
- **Solution**
- (a) Ground Area = 175 x 250 x 2000 x 2000 mm^2

  = 175 x 250 x(2000/1000)x(2000/1000) m^2

  =175,000 m^2
- Ground Area = (175 x 250) x 500^2 mm^2

  = 175 x 250 x(500/1000)x(500/1000)m^2

  = 10937.5 m^2

- **Example 5**

  A plot of land was surveyed and found to have an area of 2000 m^2. If it is plotted on a plan 1:500, what will be the plan area in mm^2?

Answer = 8000 mm^2.

- **Example 6**
- A plot of land is in the form of a rectangle in which the length is twice the breadth. When surveyed, it was found to have an area of 16,722.54 m^2. Calculate the length of the sides as drawn on a plan whose scale is 1: 10 560. **ANSWER = 8.66 mm x 17.32 mm**

Distance measurements

- Distances can be measured **directly** with the (tape, rod, etc.)

- Distances can be measured **indirectly with** optical or electronical methods. Theodolite, Total Station, GPS

Types of Distance Measurement

- Depending on the **relative positions** and **elevations** of the two points invovled, the distance could be
- Horizontal
- Slope or
- Vertical.

- In a 3-D coordinate system,

i. Horizontal distance is obtained if the 2 pts have the same Z – value.

ii. Slope distance – if the 2 pts have different values for all X, Y and Z coordinates.

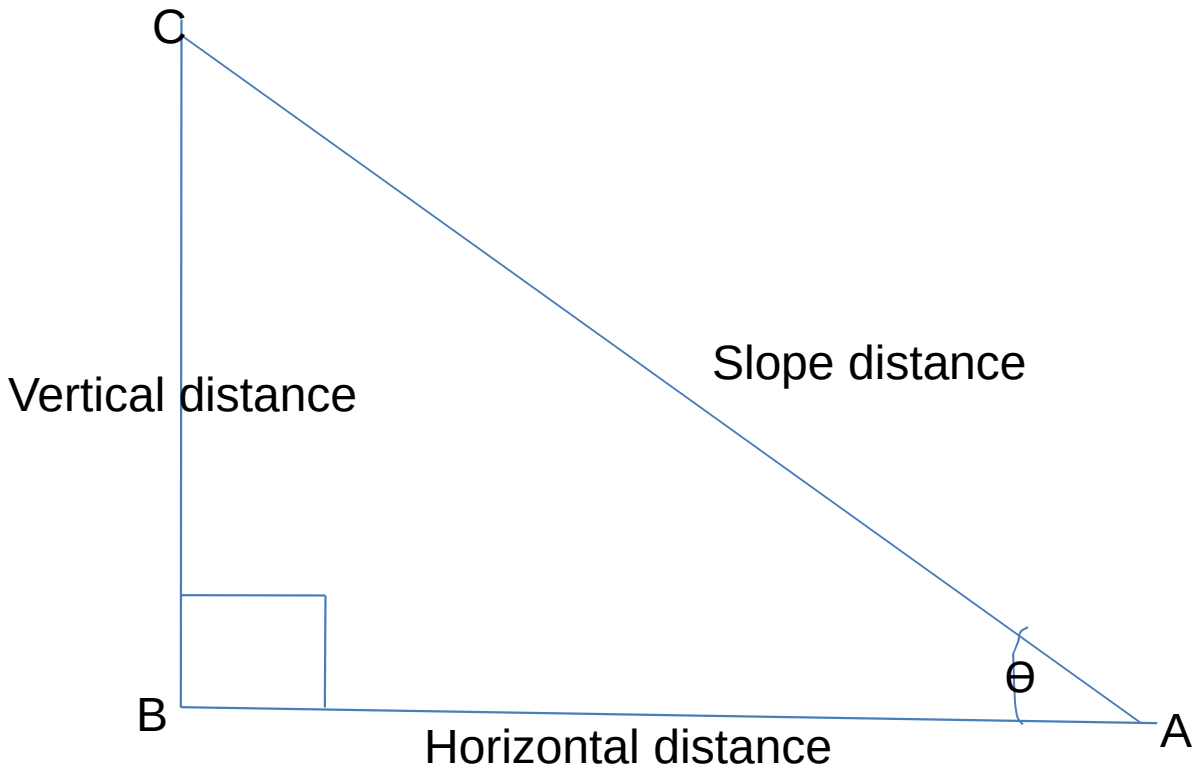
iii. Vertical distance- if 2 pts have the same X and Y coordinates but different Z coordinate

In plane surveying, horizontal distances are needed thus the distance between two points are reduced to its equivalent horizontal distances by

i. The method used or

ii. from computation - slope distance-horizontal dist.

Types of distance measurement

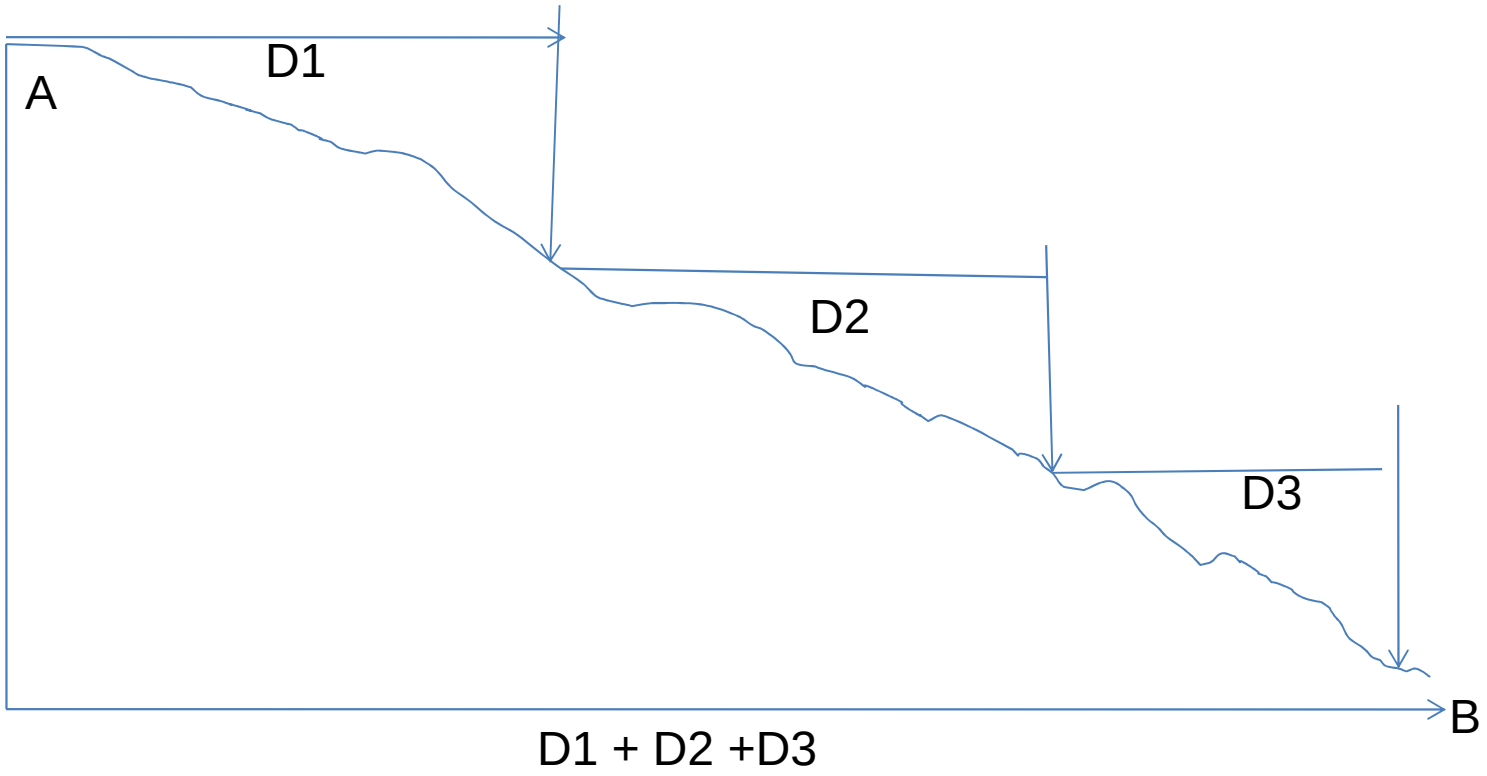


Direct measurement

- Taping is a direct means of determining the straight line distance between two points using a tape. (Field work)
- Pacing etc

Step chaining

- Horizontal distance measurements on sloping ground.



Measurement of slope distance

- Abney level (explain)
- **Examples of slope correction**

1. Calculate the plan length for a measurement of 126.3metres along a gradient of 2° 34′.

- **Solution**

Plan length = slope length X cosine inclination

= 126.3 X Cosine 2° 34′

= 126.3 x 0.9990

= <u>126.174m</u>

- 2. Calculate the plan length where a distance has been measured along a slope of 1 in 3 and found to be 149.5m.

Solution

- Plan length = slope length X cosine inclination

$$= \frac{149.5 \times 3}{\sqrt{10}}$$

$$= \frac{149.5 \times 3}{3.162} = 141.840\text{m}$$

Reduction of slope measurements by difference in elevation

- Pythagoras theorem- if the differences in elevation between the two ends of the tape (d) have been measured by levelling.

$H = [L2 - d2]1/2$

Example 1.

A distance of 290.430 m was measured along a smooth slope from A to B. The elevations of A and B were measured and found to be 865.2 and 891.4 m, respectively. What is the horizontal distance from A and B?

Errors in making measurements

- An error is the difference between a measured value of a quantity and its true value. That is

E = M-T, Where E is the error in a measurement, M the measured value, and T is the true value.

It can be unconditionally stated that:

i. No measurement is exact
ii. Every measurement contains errors
iii. The true value of a measurement is never known; and therefore
iv. The exact error present is always unknown.

Mistakes

- These are observer blunders and are usually caused by a misunderstanding of the problem,

Carelessness, fatigue, missed communication, or poor judgment. Examples are transposition of numbers such as recording 73.96 as 79.36; failure of a full tape length.

- These can be detected by systematic checking of all work and eliminated by redoing part of the job or even all of it.

Sources of Errors

- There are basically three main sources of errors, namely, natural, instrumental and personal.

- Natural sources of error may be caused by adverse or variable weather conditions. Examples of which are; expansion or contraction of steel tapes due to temperature changes, refraction of light rays, variation in the speed of electromagnetic waves through the atmosphere, etc.

- Instrumental errors are caused by imperfection and adjustment of equipment used in surveying.

  The effect of many instrumental errors can be reduced, or eliminated, by applying computed corrections or adoption of proper surveying procedures.
- Personal errors results from the inability of the individual to make exact observations due to limitations of human sight, touch and hearing.

Classification of Errors

- Two classes- Systematic and Random

Systematic errors are attributed to known circumstances and can be constant (having the same sign and magnitude) or variable throughout an operation.

Repeated observations will increase the effect of systematic errors and are the most difficult to deal with.

They always recur in the same instrument or operation and require very careful consideration prior to, during and after the survey.

- Systematic errors are minimised by systematic observation and reduction techniques.
- Examples are refraction of light rays, expansion or contraction of steel tapes due to temperature variation,
- misalignment of the tape, error due to sag of the chain in stepping operations, the chain not being true to length and the chain not being pulled straight etc.
- These are Standardisation, temperature, Tension and slope.

Random Errors

- Random errors are associated with the skill and vigilance of the surveyor.
- They are introduced into each measurement mainly because no human can perform perfectly. These are caused by variations in the eye sight of persons using the instrument and sadden changes of temperature or wind and atmospheric disturbances.
- They are usually small and not cumulative.Thus equipments and our observations are imperfect. They represent the residual error after all other errors have been eliminated.

- Taking the mean of repeated observations minimise their effects.
- The characteristics of random errors are as follows:

i. Small errors occur more frequently than large ones

Ii Positive and negative errors of equal magnitude occur with equal frequency; that is they are equally probable.

iii. Large errors are rare and are more likely to be mistakes or untreated systematic errors.

Precision and Accuracy

- Accurate and precision are often used indiscriminately to describe the quality or reliability of measurements or observations as if they were synonymous.
- *Accuracy is the nearness of measured value to the true value of a quantity.* Since the true value is seldom known, accuracy is generally indeterminate in practice.

- *Precision is an indication of the spread of the measured values of a quantity.* Several values grouped closely together are said to constitute a more precise set of measurements than one with a broader spread of values.
- Example
- Two groups of students measured a line of 100 m nominal length with a tape and obtained the following results:

Group A: 99.01, 98.99, 99.02, and 98.98m

Group B: 99.85, 99.90, 100.15, and 100.00 m

- group A precise measurement and Group B is accurate

Applying corrections to Tape Measurements

- Standardisation correction: An error due to incorrect length of a tape. Standardisation length different from True length.
- $CL = \left[\frac{AL - NL}{NL} \right] \times ML$

And $TL = ML + CL$

$= \frac{\{AL\}ML}{NL}$

- Where CL = correction to be applied to the measured length to obtained true length;
- AL = Actual length (Standardisation)

NL = Nominal length (Given by manufacturers)

ML = Measured length of line

TL = Corrected (or true) length of line

A 100 m steel tape when compared with a standard is actually 100.020 m long. What is the correct length of the line measured with this tape and recorded to be 565.750 m?

Solution

- NL = 100.00 m

AL = 100.02 m

$$CL = \left[\frac{100.020 - 100.000}{100.000}\right] 565.75$$

$$= + 0.113 \text{ m}$$

Corrected Length, TL = ML +CL = 565.75 + 0.113

= 565.863 m

- Note:

i) AL –NL = amount by which a full tape is too long or too short = 0.02 m

ML/NL = number of tape lengths in the measured line = 5.6575

CL = 5.6575 * 0.020 m = 0.113 m

ii) In measuring unknown distances with a tape that is too long, a correction must be added. Conversely, if the tape is too short, the correction will be minus, resulting in a decrease.

- A canal to be constructed was in a form of a straight line ABCD. It was measured in 3 sections along the slope. The angles were measured by a theodolite and the ff. observations obtained.

Line	length(m)	Angle of slope
AB	205.00	+6° 30’
BC	115.90	+2° 30’
CD	96.40	-5° 00’

Determine the plan length of the line AD and calculate the gradient of the canal to be made between points A and D.

Changes in temperature: Temperature correction

- When field temperature differs from standard temperature of tape, a temperature correction must be computed and applied to the measured length.
- Correction $CL = L\alpha(Tm - Ts)$

and $Lc = L + Ct$ Where:

Ct = temp. correction to be applied

L = Length of actual tape

α= Coefficient of linear expansion

Tm =Tape temp. at time of measurement

Ts = Tape temp. when it has standard length.

Tape not horizontal: slope correction

- If tape is not horizontal an error is introduced. The amount of error is computed as follows:

$$C_s = -L_s(1 - \cos\phi)$$

$$= -d^2 / 2L_s$$

Where $d = \Delta h$ = difference in height between the ends of the line.

L_s = measured slope distance

ϕ = slope angle

Indirect Distance Measurements

- An indirect measurement is made when it is not possible to measure the distance directly. The required distance is measured by its relationship to some other measured values.
- Stadia Tacheometry- tacheometry is a method used to quickly determine the horizontal distance to, and elevation of a point.
- It is used to determine the position and height of points from which contours are interpolated and also for the positions of points of detail during the production of large and medium scale topographic maps/plans.

- It can also be used to provide horizontal and vertical controls. Instruments required for stadia measurements are:

i. A theodolite or level instrument with stadia hairs (upper, middle and lower)
ii. A levelling staff
iii. A tape

Measurements taken are:

i. Staff intercept (upper, lower and middle)
ii. Vertical angle
iii. Centre hair reading of the staff
iv. Height of the theodolite measured using tape.

Indirect computed quantities

i. Horizontal distance between two stations, P and X.

$$H = Ks\cos^2\theta + C\cos\theta$$

ii. The vertical component (V) of the inclined line of sight given by:

$$V = \tfrac{1}{2}(Ks \sin 2\theta) + C \sin\theta$$

iii. The reduced level of the staff position:

$$RLx = RLp + hi \pm V - m$$

- Where k = multiplying constant of the telescope, usually set at 100;
- S = stadia interval on the staff = upper – lower
- m = centre hair reading of the staff at x;
- θ = vertical angle
- hi = height of the theodolite at p;
- C = additive constant, usually taken to be zero and
- RLp, RLx = reduced level of stations P and X, respectively.

ANGULAR MEASUREMENTS

INTRODUCTION

INTRODUCTION

- Angular measurements are carried out for

i. Orientation of lines and

ii. Determination of locations of points

Measured directly in the field by:

i. Digital Theodolite

ii. Total stations

iii. Optical Theodolites

iv. Compasses

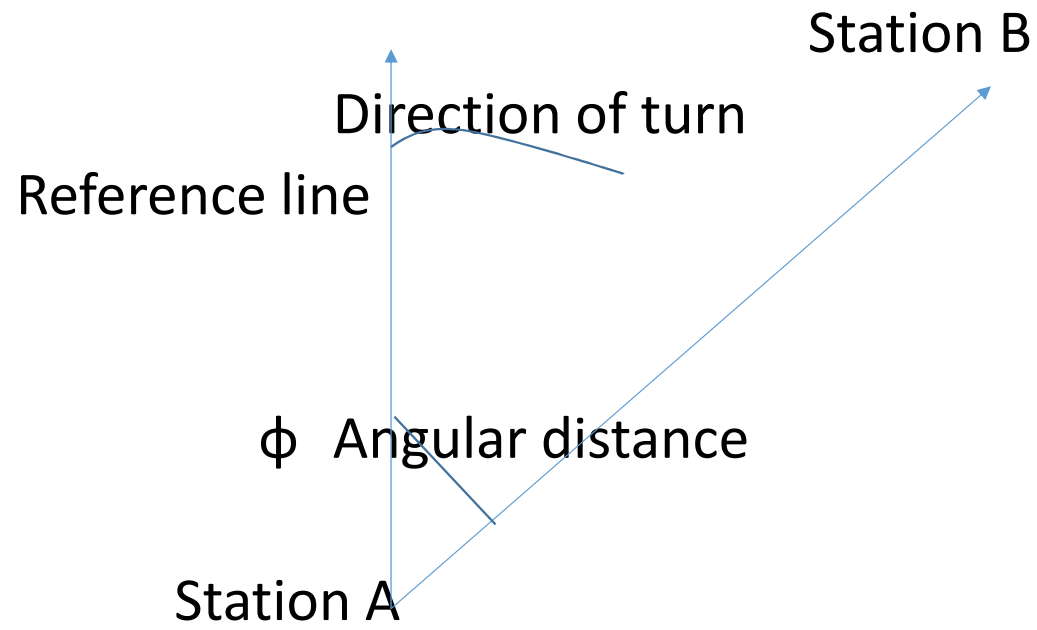
- To measure an angle the following basic requirements are needed:

i. Reference / starting line;

ii. Direction of turning;

iii. Angular distance (i.e. value of the angle)

To reference object (RO)



Basic requirements in determining an angle

Measurement of Horizontal Angle

- Angles are classified depending on the plane in which it was measured:

i. Horizontal or

ii. Vertical angles

A horizontal angle is the angle formed in a horizontal plane by two lines extending from the same source or point.

- Horizontal angle has a direction; It is either clockwise (to the right) or anticlockwise (to the left).

- Horizontal angles are the basic measurements needed for determining bearings and azimuth (angular measurement in spherical coordinates)

Different types of horizontal angles

Internal and External angles

- Internal angles can be clockwise if the direction of turn is clockwise but anticlockwise if direction of turn is anticlockwise.

- External angles are measured clockwise if direction of turn is clockwise

Theodolite Surveying

- Theodolites are precision instruments used extensively for measuring angles in the horizontal and vertical planes.

- Many different theodolites are available for measuring angles and they are classified according to the smallest reading that can be taken with the instrument.

- For example, a 1” theodolite is one which can be read to 1” directly without estimation.

The components of the theodolite





Description of a Theodolite

- Precision instruments used extensively in construction work for measuring angles in both the horizontal and vertical planes.
- We have different types of theodolites but are classified according to the smallest reading that can be read with the instrument. Eg a 1" theodolite is one that reads up to 1" directly without estimation

- Vertical circle:
- Horizontal circle
- Telescope
- Eyepiece
- Circle reading eyepiece.
- Reflecting mirror for the reading system
- Plate level
- Circular bubble
- Upper plate (or alidade): it carries the horizontal circle reading system.
- Lower plate: this carries the horizontal circle.
- Tribrach
- Levelling footscrews
- Optical plummet
- Telescope clamp

Setting up the Theodolite

- Set up the tripod
- Mount the theodolite on tripod
- Centre the theodolite over the ground mark (optical plummet or plumb line)
- Level the theodolite by centralizing the horizontal plate bubble and
- Remove parallax
- Face left (L) : vertical circle reading to the left of observer
- Face right (R): vertical circle reading to the right of observer
- LL meaning face left, swing left
- RR meaning face right, swing right
- Plunging/transiting the telescope ie rotating the telescope about the transverse axis

Booking and calculating angles

Instrument Station	Face & Swing	Back Station	Fore Station	Horizontal circle reading	Included Angle	Mean Included Angle
B	LL	A		00°00'00"		65°25'10"
	LL		C	65°25'00"	65°25'00"	
	RR		C	245°25'20"	65°25'20"	
	RR	A		180°25'00"		
B	LL	A		20°00'00"		65°25'10"
	LL		C	85°25'00"	65°25'00"	
	RR		C	265°25'20"	65°25'20"	
	RR	A		200°00'00"		

Measuring Interior angles

- The sum of the interior angles in a closed figure is (n-2)180, in which n is the number of sides.
- For external angles, (N +2)180.
- Note
- If stations are occupied in a clockwise direction, the interior angles are measured anticlockwise direction. If the stations are occupied in an anticlockwise direction, the interior angles are in a clockwise direction.

Example

Station	Observed angle
A	115°11'20"
B	95°00'20"
C	129°49'20"
D	130°36'20"
E	110°30'00"
F	138°54'40"
Sum	720°02'00"

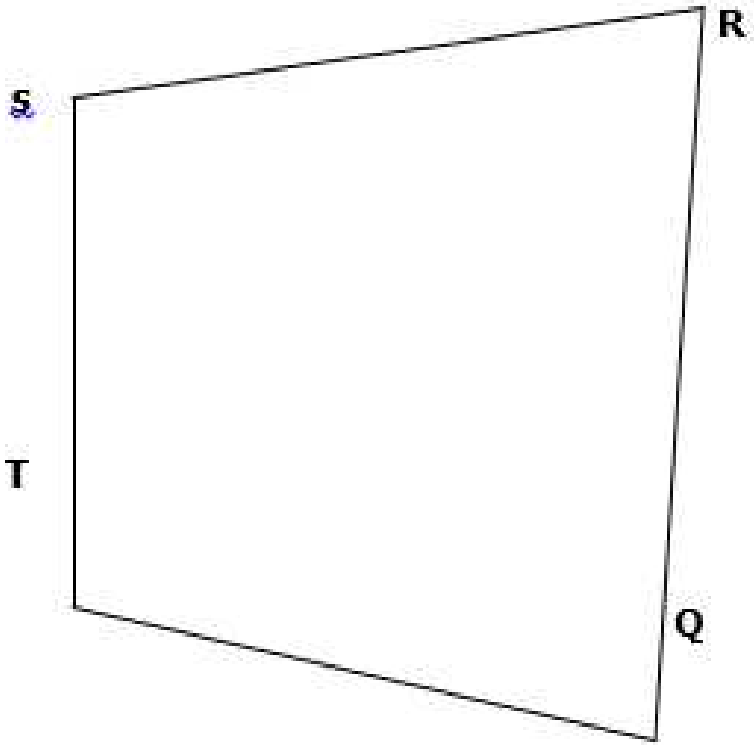
Required sum = (2*n* 4) x 90° = (2 x 6 4) x 90° = 720°00'00"

Hence misclosure = 720°02'00" - 720°00'00" = + 20'00"

Adjustment of angles

- The angles are adjusted by distributing the misclosure over all the measured angles.
- For the above the adjustment per angle will be – (20'00'')/6, since six (6) internal angles were measured

Station	Observed angle	Adjustment	Adjusted angle
A	115°11'20"	-20"	115°11'00"
B	95°00'20"	-20"	95°00'00"
C	129°49'20"	-20"	129°49'00"
D	130°36'20"	-20"	130°36'00"
E	110°30'00"	-20"	110°29'40"
F	138°54'40"	-20"	138°54'20"
Sum	720°02'00"	-20'00"	720°00'00"



Angle	Observed value
TSR	240° 17′ 01″
SRQ	279° 40′ 56″
RQT	300° 18′ 52″
QTS	259° 42′ 55″

Adjustment of Theodolite Surveys

Angular misclosure

- The first step is to determine the angular misclosure of traverse. That is, to check that the observed theodolite angles sum to their required value

- ***Check for polygon traverse***

The observed angles of a polygon traverse can be either internal or external angles.

For internal angles: Sum of internal angles = (2*n* – 4) x 90°

For external angles: Sum of external angles = (2*n* + 4) x 90°

- Where *n* is the number of angles measured

Check for link traverse

Sum of angles = (final forward bearing – initial back bearing) + *m* x 180°

- Where *m* depends on the nature of the traverse. *m* = (*n* – 1)

  or (*n* + 1),

- where n is the number of angles measured between the initial back bearing and final forward bearing

Measuring with the theodolite

Faces of the theodolite

- Face Left (L)
- Face Right (R)

The theodolite can be swung in two directions:

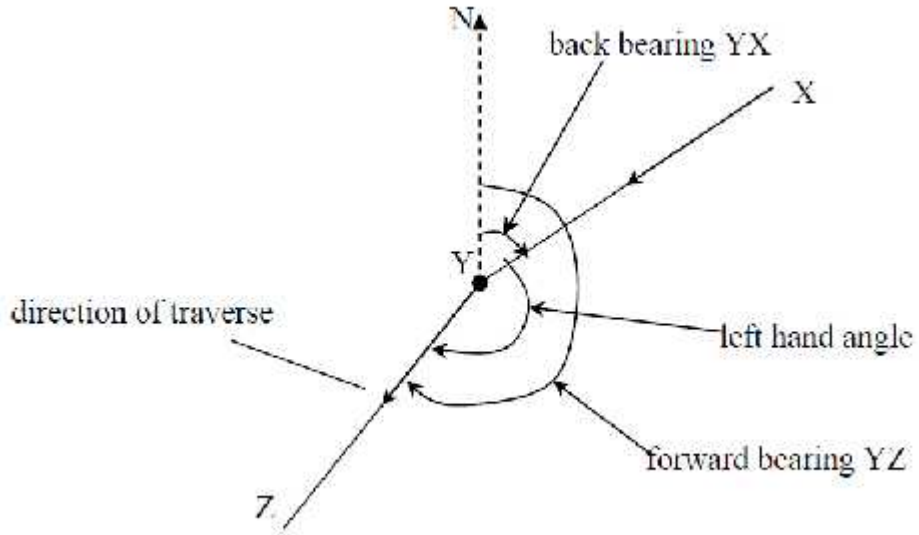
- Swing Left (L) or anticlockwise
- Swing Right (R) or clockwise

Combining the face and swing we can have:

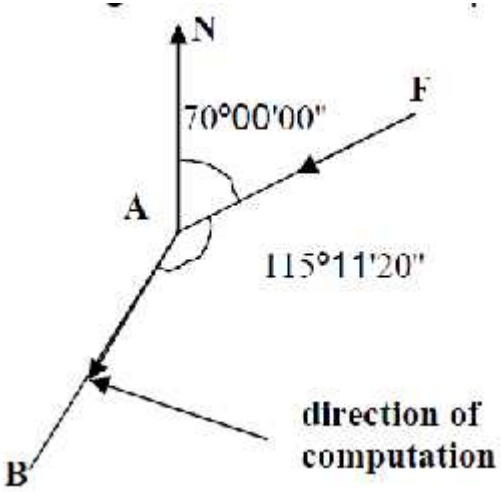
- **LL** meaning face left, swing left
- **RR** meaning face right, swing right

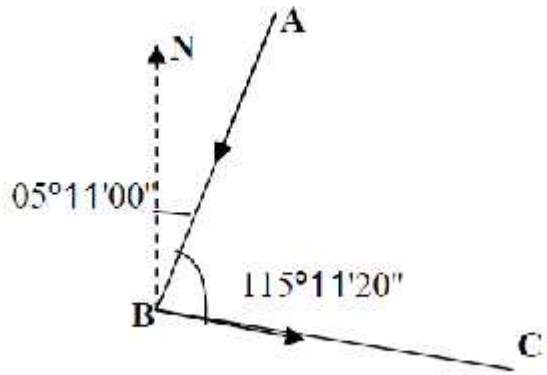
Conversion of angles to whole circle bearings

- This is done because the angles measured by the theodolite are not bearings



Forward bearing = back bearing + left-hand included angle





At station A:

forward bearing AB = back bearing AF + left-hand angle at A
− 70°00'00" (given) + 115°11'00"
− 185°11'00"

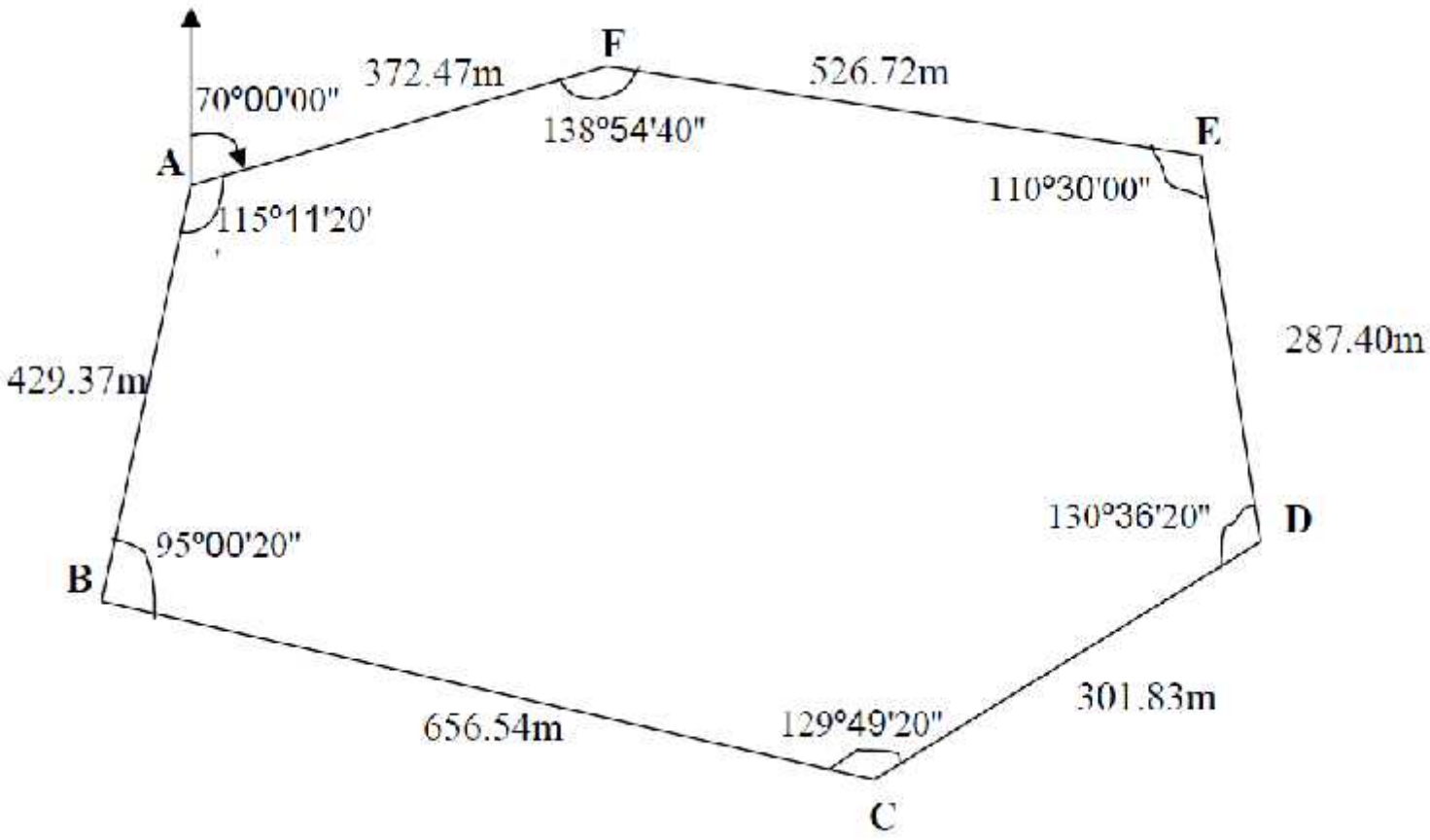
At station B:

forward bearing BC − back bearing BA + left-hand angle at B

but back bearing BA − forward bearing AB ± 180°
= 185°11'00" ± 180°
− 365°11'00" or 05°11'00"
− 05°11'00" (to keep bearing in the range 0 to 360°)

hence forward bearing BC − 05°11'00" + 95°00'00"
− 100°11'00"

Final drawing



Meridians

- Four types
- Namely

i. Astronomical/ Geographical/True meridian
ii. Magnetic meridian
iii. Grid meridian
iv. Assumed meridian

- True meridian

It is an imaginary line on the mean surface of the earth joining the north and south poles. This meridian is fixed.

- Magnetic Meridian at a point

It is a line passing through the point and joining the earth's magnetic north and south poles. Magnetic meridian is not fixed. Deviation from the True meridian is called Magnetic Declination. Declination is + / positive if it is right of the True meridian and negative if it is to the left.

- Assumed meridian

Is any arbitrary line of a survey chosen as a reference line for convenience in a survey of limited area.

- Grid Meridian

Lines that are parallel to the central meridian chosen as a reference line in a map projection system of a country. In Ghana, Longitude 1° West and all grid meridians are parallel to it.

Bearing of a line

- The bearing of a line AB is defined as the clockwise horizontal angle measured from the meridian passing A to the vertical plane containing the line.

- The bearing of a line gives us the direction of that line with respect to a reference meridian.

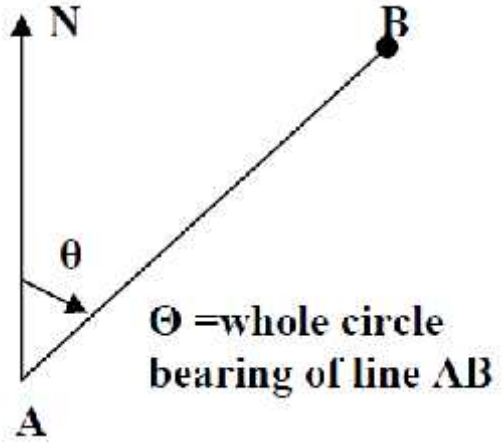
- Bearing is always measured from the north for countries in the northern hemisphere, Ghana.
- Bearings may be of different types depending on the type of reference meridian:

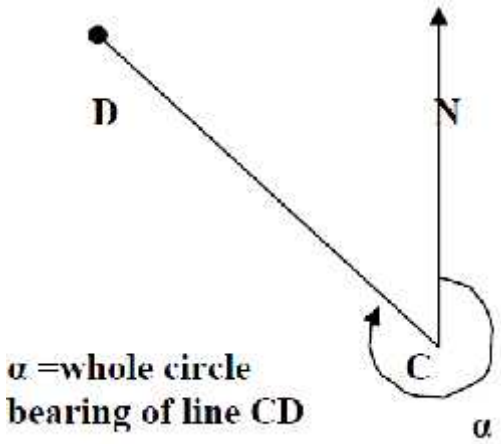
- Magnetic bearing is with respect to magnetic meridian.

Whole-circle Bearings

- To establish the direction of a line between two points on the ground, its bearing has to be determined.

- The whole-circle bearing (WCB) of a line is measured in a clockwise direction in the range 0° to 360° from a specified reference or north direction





Quadrant bearing (reduced bearing)

- If the cardinal point of a compass are drawn and labelled north , east, south and west moving clockwise, the whole 360° circle would have been divided into four quadrants of 90°

- *The quadrant bearing of any line is the angle it makes with the north-south axis.*

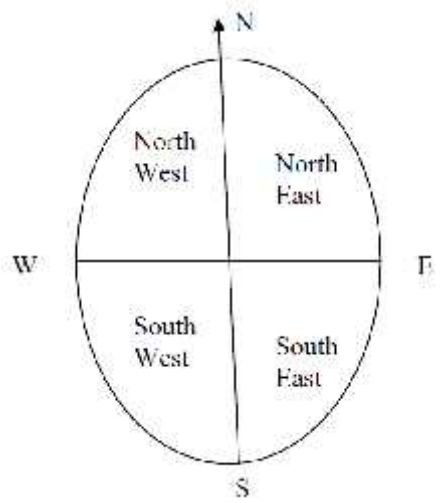
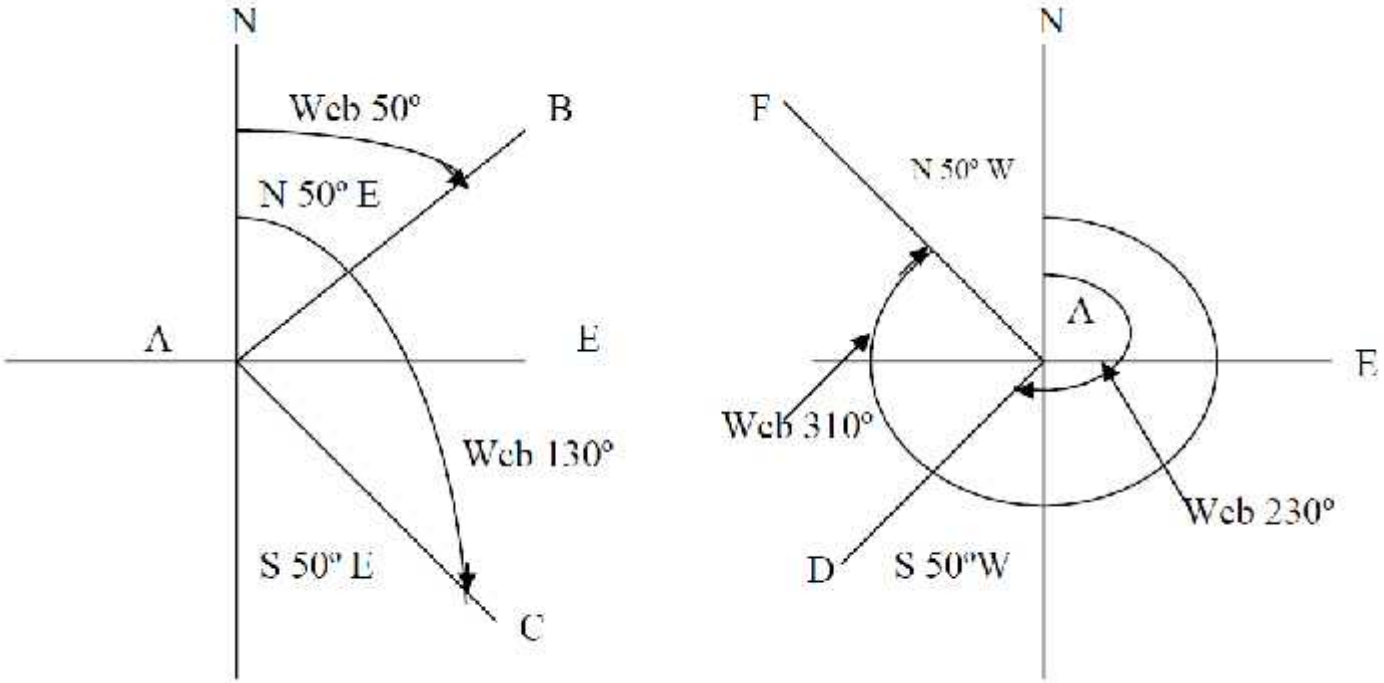


Illustration of Quadrant and Whole circle bearings



To convert quadrant to whole-circle bearing, the following rules are applied:

a. If the bearing is in the 1st quadrant (N-E), no change is made
b. If the bearing is in the 2nd quadrant (S-E), subtract it from 180°
c. If the bearing is in the 3rd quadrant (S-W), add 180° to it.
d. If the bearing is in the 4th quadrant (N-W), subtract it from 360°

To reduce whole-circle bearings, the following rule is applied:

a. If the bearing lies between 0° and 90°, no reduction is required.
b. If the bearing lies between 90° and 180°, subtract it from 180°
c. If the bearing lies between 180° and 270°, subtract 180° from it.
d. If the bearing lies between 270° and 360o, subtract it from 360°

Examples:

- With the aid of a diagram, find the quadrant bearing for the following whole circle bearing (wcb)
  - 310^0
  - 45^0
  - 225^0

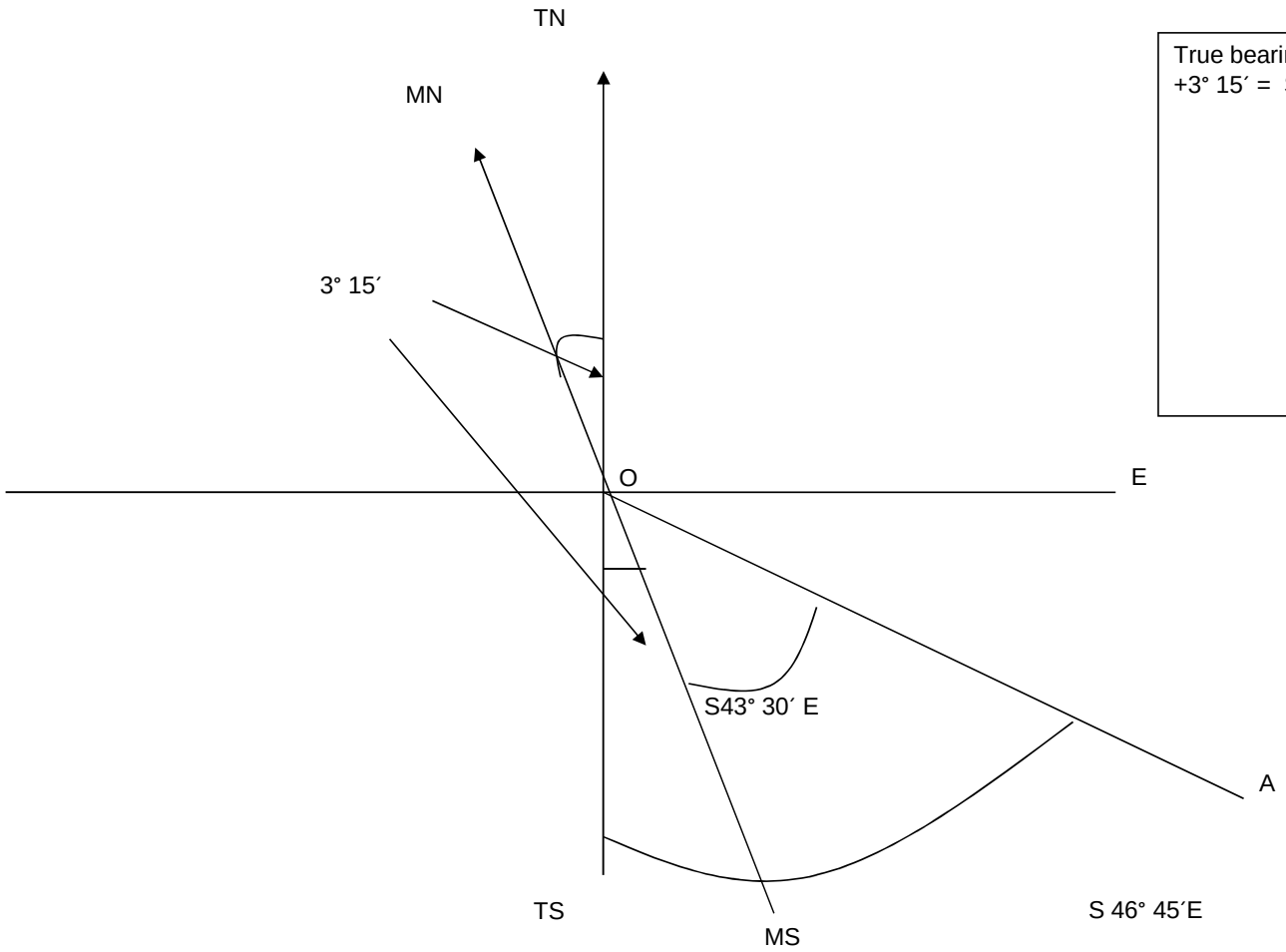
- The magnetic bearing of a property line was S43^o 30´ E. The magnetic declination at the place of the survey was 3° 15´ W of true North. What is the true bearing of the line?

- Whole Circle Bearing
- Quadrant bearing

- If the whole circle lies between 0° and 90°, the quadrant bearing has the same numerical value in the North east quadrant.
- If the wcb lies between 90° and 180°, the quadrant bearing is (180 – wcb) and lies in the south east quadrant.
- If the wcb lies between 180° and 270°, the quadrant bearing is (wcb - 180) and lies in the south west quadrant.
- If the wcb lies between 270° and 360°, the quadrant bearing is (360 - wcb) and lies in the north west quadrant.

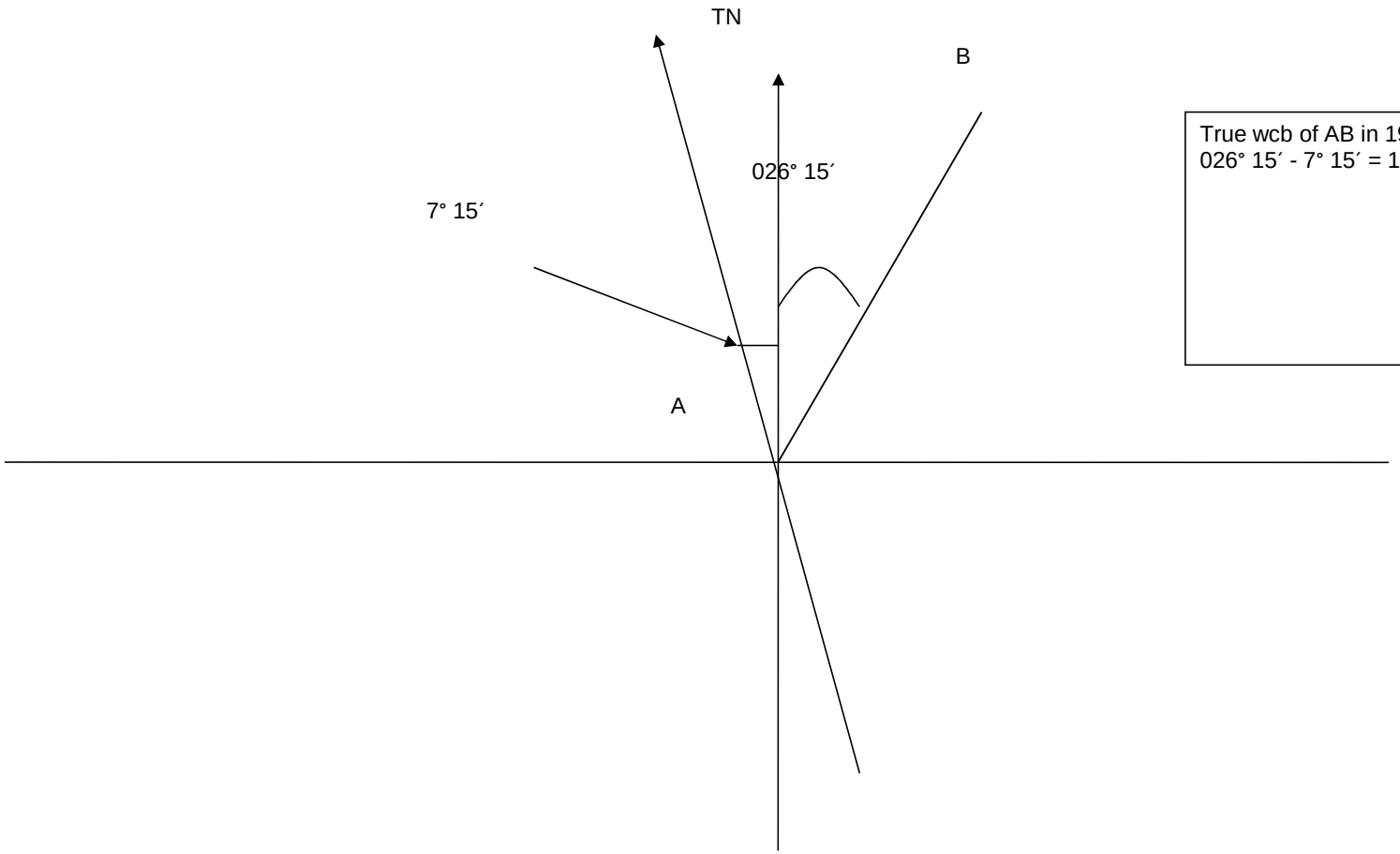
Questions

- The magnetic bearing of a property line was S43° 30´ E. The magnetic declination at the place of the survey was 3° 15´ W of true North. What is the true bearing of the line?
- The magnetic wcb of a line AB read in 1900 was 026° 15´, the declination at that time was 7° 15´W, and in 2006 the declination is 4° 30´E. What is the magnetic bearing of AB in 2006?



True bearing of OA is given by: S43° 30´ E
+3° 15´ = S 46° 45´E

Fig A



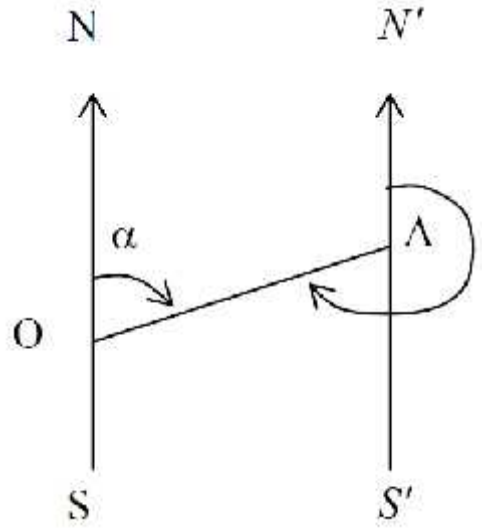
True wcb of AB in 1900
026° 15′ - 7° 15′ = 19° 00′

Fig. B

Forward and back bearing

- Every line has two bearings, *the forward and back bearings.*
- The forward and back bearing of a line differ by 180° if there is no local attraction in the case of compass bearing
- The bearing of a line, designated by the stations between which it lies, is taken as that from the first station mentioned.
- Thus, bearing of a line OA, it is the direction of the line from O to A.
- The direction from A to O is termed the ***back, or reverse, bearing.***

- The bearing of OA, that is the direction from O to A with reference to the grid line meridian NOS, is the clockwise angle N$\hat{O}$A=α.
- The bearing of AO, i.e., the back bearing of OA is the clockwise angle $N'\hat{A}O = N'\hat{A}S' + O\hat{A}S'$
- Since $N'\hat{A}S'$ is a straight line, $N'\hat{A}S' = 180^o$ and $O\hat{A}S' = N\hat{O}A = \alpha$ (alternate angles)



NOTE

- If the forward bearing is less than 180° add 180° and if it is more than 180° subtract from 180° to get the back bearing
- What happens in the quadrantal system?

Back bearing of a line

- **Every line has two bearings, the forward and back bearings. The forward and back bearing of a line differ** by 180 ° if there is no local attraction in the case of compass bearing.

- Find the back bearings of the following:

i. S 15 56 30 E ii. N 85 40 30 W iii. S 65 35 25 W

Local Attraction

- The free flowing , well balanced magnetic needle will not come to rest in the magnetic meridian, if there is any material such as mineral deposits, particularly magnetite ($Fe_3 O_4$), in the locality.
- The proximity of steel structures, rails, current-carrying electric cables, iron fences and pipes, lamp posts, machetes and felling axes and even, chains and bunches of keys in your pockets may produce undesirable interference.
- Thus, a bearing read from a compass under the influence of such attractive forces in the vicinity, may originate from a direction other than that of the magnetic meridian, and the error in the bearing is proportional to the magnitude of the attractive force.
- *Local attraction* is the term given to both the cause and effect of this error.

- **Detection of local attraction**
- Observe the bearing of each line in both directions. If the forward and back bearings differ by 180°, the inference is that there is no local attraction at either of the stations and that in the absence of other instrumental errors, the observation is correct so far as the reading is concerned.

- If the two bearings do not agree within permissible error of reading, check the forward bearing, since the discrepancy may have been caused by avoidable attractions such as chains, bunches of keys, machetes, felling axes, etc carried by persons around.

- If the discrepancy still exists, the presence of local attraction, at one or both stations, is suggested. (NB. An allowable error of between 1°- 2° may be a guide).

Removal of Local Attraction

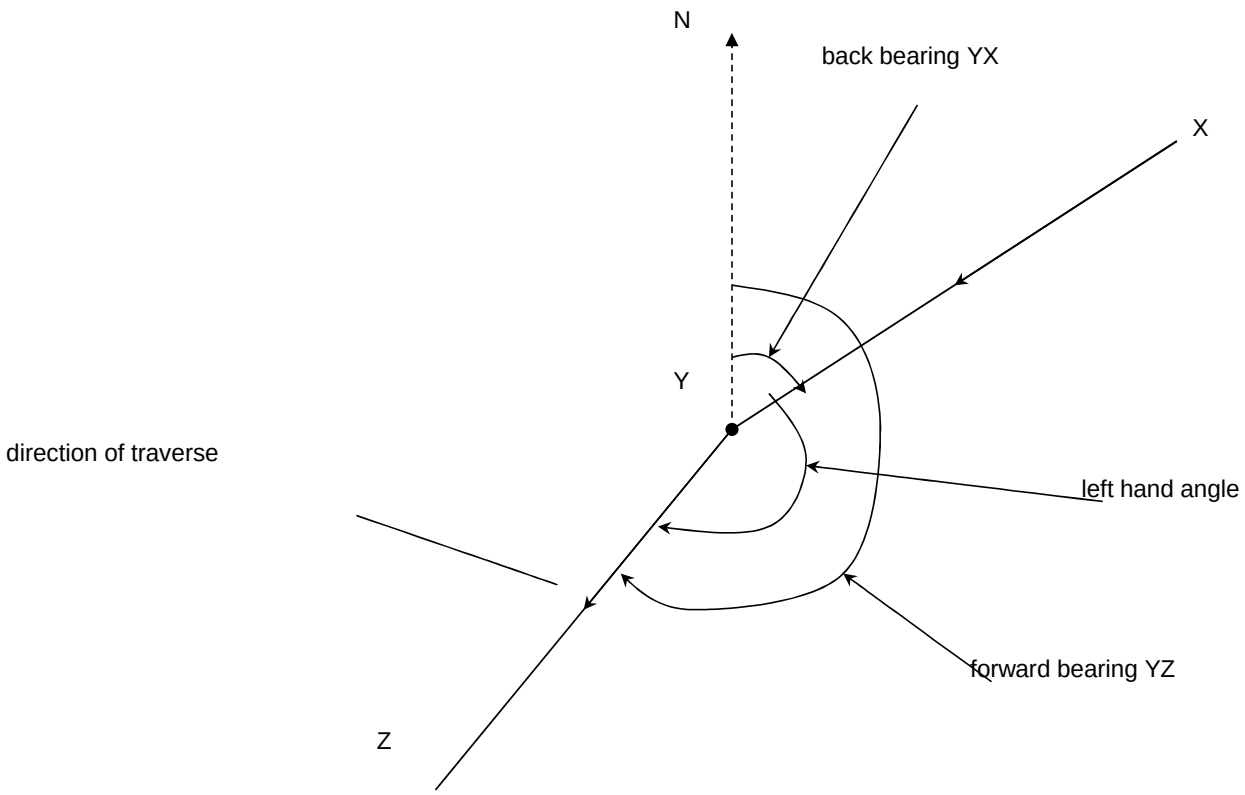
Line	Observed Bearing	Correction	Corrected Bearing
AE	319 ° 00´	-2° 30´	316 ° 30´
AB	72° 45´	-2° 30´	70° 15´
BA	252°00´	-1° 45´	250° 15´
BC	349° 00´	-1° 45´	347° 15´
CB	167° 15´	0°	167° 15´
CD	298° 30´	0°	298° 30´
DC	118° 30´	0°	118° 30´
DE	229° 00´	0°	229° 00´
ED	48° 00´	$+1^{o}$	49° 00´
EA	135° 30´	$+1^{0}$	136° 30´

Question2

The following compass traverse was carried out in the vicinity of overhead power lines.

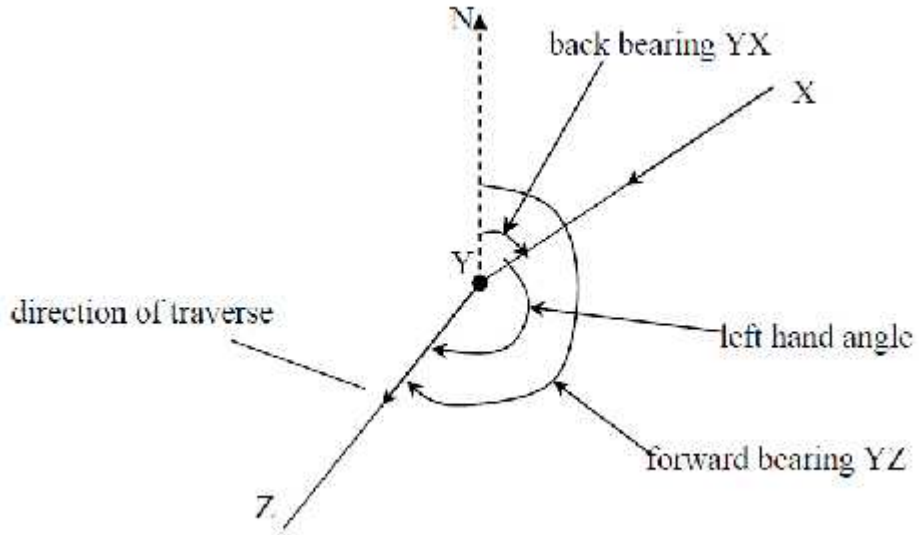
Line	W C bearing		Distance (m)
	Forward	Back	
PQ	030°	205°	32
QR	097°	272°	75
RS	164°	346°	33
ST	235°	55°	21
TU	261°	081°	88

The conversion of the angles to bearings for the theodolite surveys is illustrated in figure. This is done because the angles measured by the theodolite are not bearings.

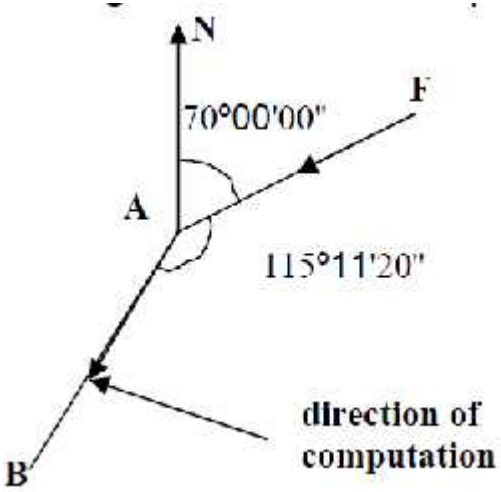


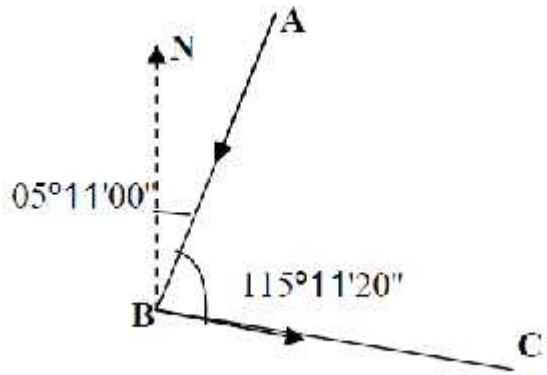
Conversion of angles to whole circle bearings

- This is done because the angles measured by the theodolite are not bearings



Forward bearing = back bearing + left-hand included angle





At station A:

forward bearing AB = back bearing AF + left-hand angle at A
− 70°00'00" (given) + 115°11'00"
− 185°11'00"

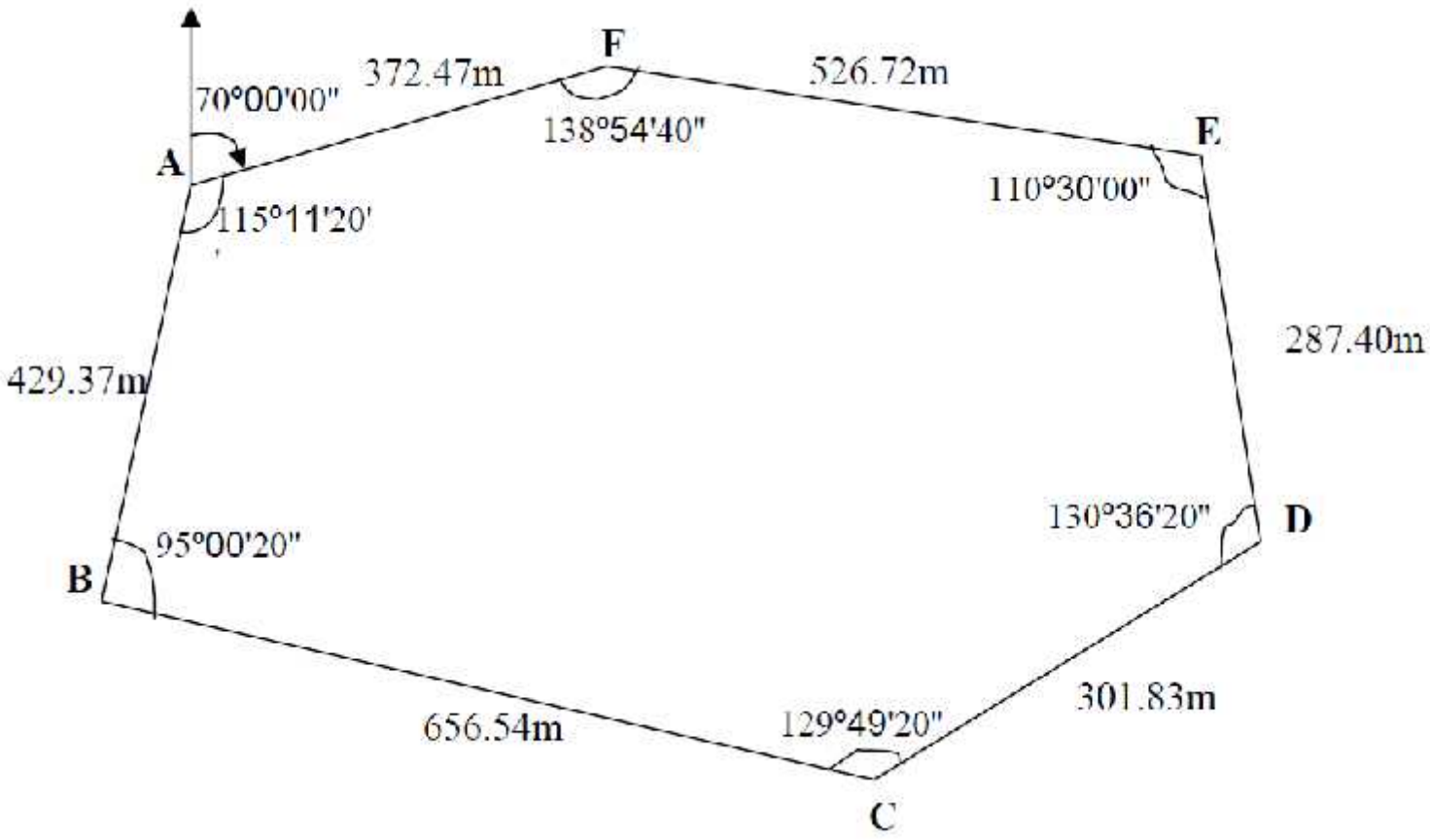
At station B:

forward bearing BC − back bearing BA + left-hand angle at B

but back bearing BA − forward bearing AB ± 180°
= 185°11'00" ± 180°
− 365°11'00" or 05°11'00"
− 05°11'00" (to keep bearing in the range 0 to 360°)

hence forward bearing BC − 05°11'00" + 95°00'00"
− 100°11'00"

Final drawing



LEVELLING AND CONTOURING

Levelling

- Levelling is the processes by which elevations/heights of points or differences in elevation between points are determined.

- Its purpose may be to;
  - provide heights or contours on a plan,
  - provide data for road sections (cross & longitudinal)
  - Calculate volumes of earthworks and other materials
  - provide a level or inclined surface in the setting out of construction works

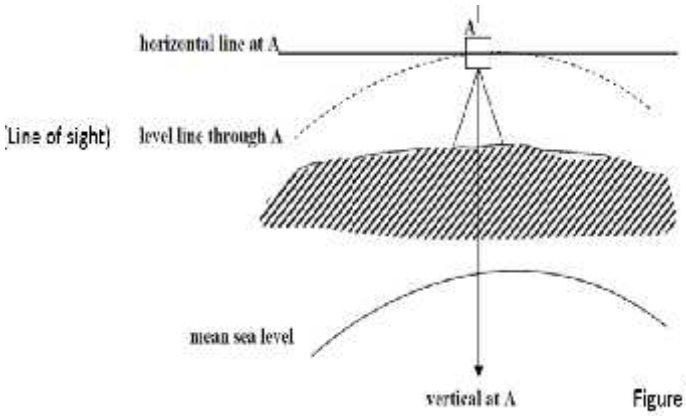
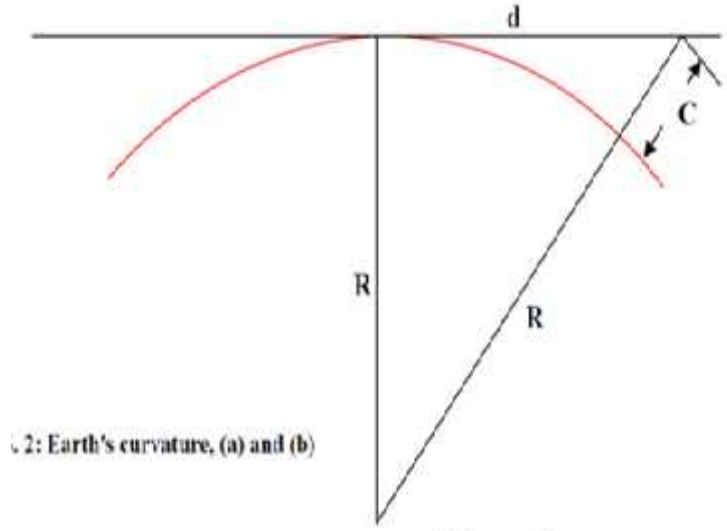


Figure 1



. 2: Earth's curvature, (a) and (b)

Figure 2

Definitions

- A ***vertical line*** is a line from the surface of the earth to the earth's centre.
- It is also referred to as a *plumb line* or a *line of gravity* because the vertical line is parallel to the direction of gravity

Horizontal line

- A horizontal line is a straight line perpendicular to a vertical line.

Level line

A level line is a curved line in a level surface.

Level surface

- A level surface is a curved surface parallel to the mean surface of the earth.
- That is, a level surface is a surface of constant elevation that is perpendicular to a plumb line at every point.

Elevation

- An elevation is a vertical distance above or below a reference datum (MSL).

Datum surface

- This is an arbitrary level surface to which the heights of all points are referred.
- This can be any surface but the most commonly used datum is the MSL or local datum point established on a construction site.

Benchmark (BM).

- A relatively permanent object, natural or artificial, having a marked point whose elevation above or below a reference datum is known or assumed.

Mean sea level (MSL)

- The average height for the surface of the seas for all stages of tide over a 19-year period as defined by the National Geodetic Vertical Datum of 1929

Curvature and refraction

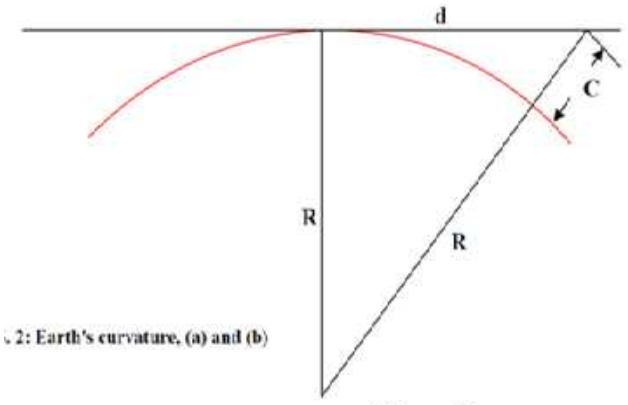


Figure 2

- From the definitions of a level surface and a horizontal line, it is evident that the horizontal plane departs from a level surface because of curvature of the Earth.

- A *level line,* is likewise a curved line and is everywhere normal to the plumb line.
- However, a *horizontal line of sight* through the surveyor's telescope is perpendicular to the plumb line only at the point of observation
- The deviation between the horizontal and the level lines is called the curvature. (shown as 'C' in figure 2)
- The curvature of the earth causes readings on the level staff to be too high by the length 'C'

Refraction

- Light rays passing through the Earth's atmosphere are bent or refracted toward the Earth's surface.
- This bending causes the reading on a level staff to be diminished or too small
- Displacement resulting from refraction is variable.
- It depends on;
  - atmospheric conditions,
  - length of line, and
  - the angle a sight line makes with the vertical

- Although the combined effects of curvature and refraction produce staff readings that are slightly too large, proper field procedures in differential leveling can practically eliminate the error due to these causes.

- For shorter distances of less than 120m these errors are negligible

- For longer distances, the effect of curvature and refraction will cancel out if the sight lengths are equal (BS = FS)

The Level Instrument

- Instruments used for differential leveling can be classified into four categories:
  - *Dumpy levels*
  - *Tilting levels*
  - *Automatic levels and*
  - *Digital levels*

- Although each differs somewhat in design, all have two common components:

  - a *telescope* to create (a line of sight and enable a reading to be taken on a graduated staff) and
  - system to orient the line of sight in a horizontal plane

Automatic level

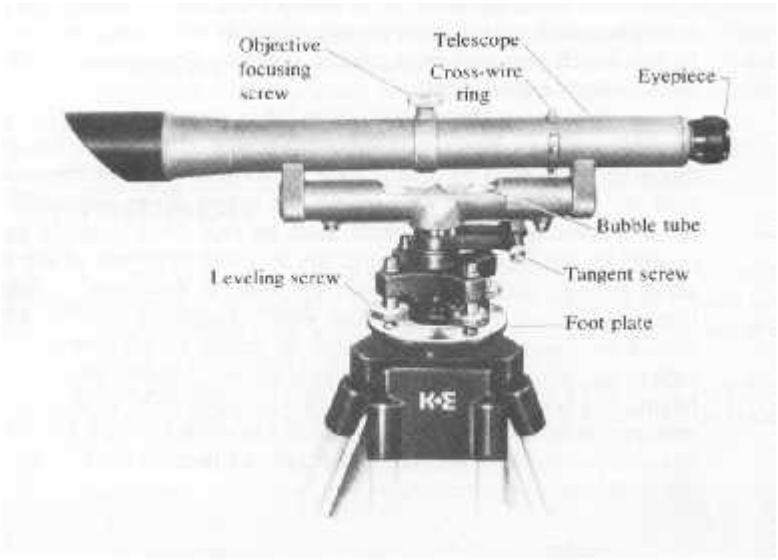


Tilting level





Digital level



Dumpy level

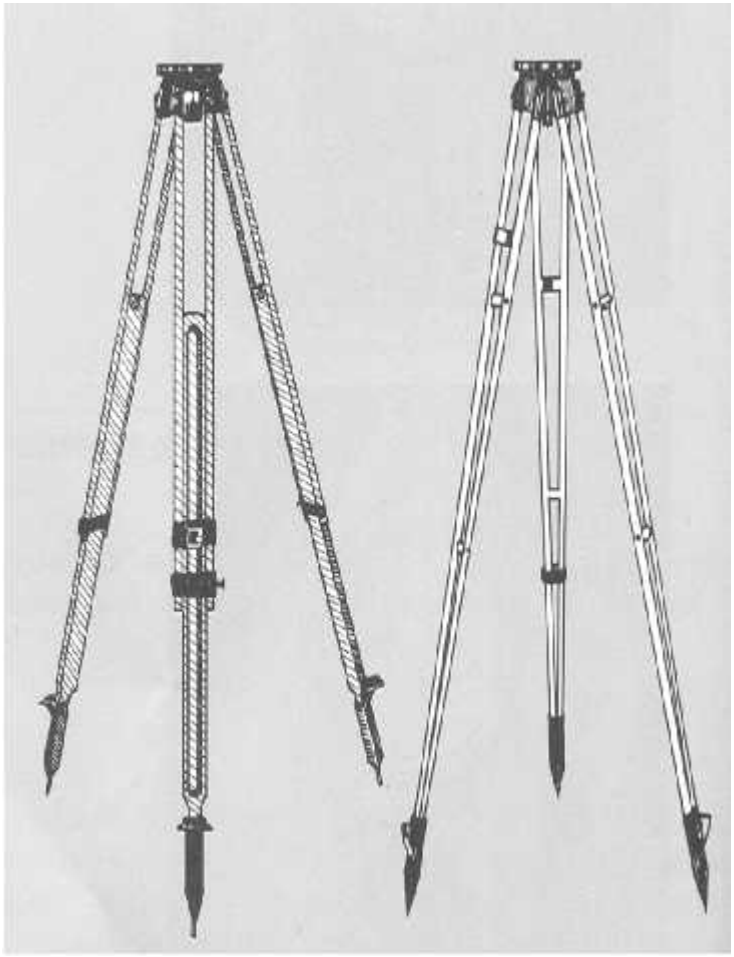
Parts of a level instrument

- Eyepiece
- Levelling screws (foot screws)
- Objective lens focus (focusing screw)
- Horizontal motion screw
- Circular or Tubular bubble
- Objective lens
- Collimator or sight
- Eyepiece focus screw
- Tilting screw

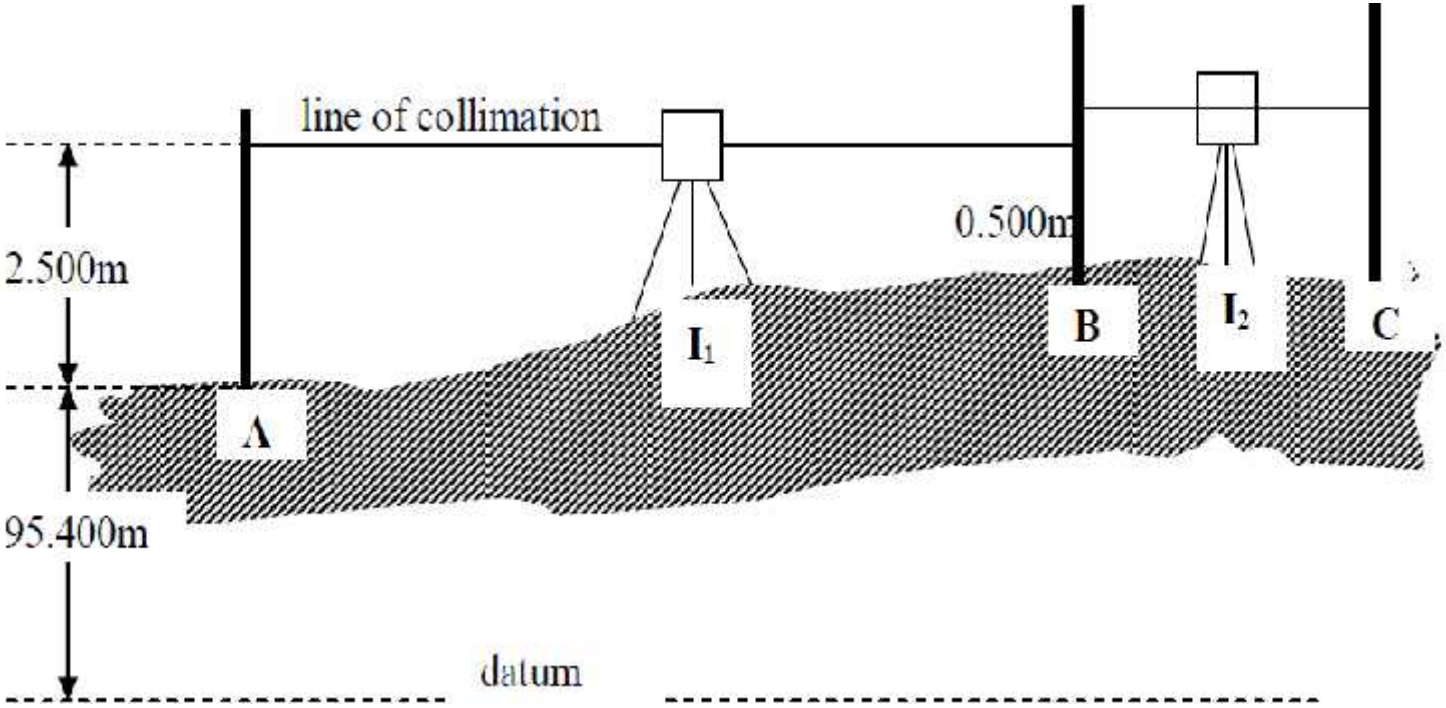
The levelling Staff and Tripod







Field procedure in levelling between two points



Explanation to Diagram

- **A** is the *back sight* **(BS)** and **B** is the *fore sight* **(FS)** when the level instrument is at $\mathbf{I_1}$
- **B** becomes the *back sight* and **C** the *fore sight* when the level instrument is at $\mathbf{I_2}$**.**
- **B** is called a *change point* (CP) since at point **B** both back sight and fore sight readings were taken.
- The **BS** is the first reading taken after the instrument has been set up.
- The **FS** is the last reading taken before the instrument is moved.
- Any reading(s) taken between the **BS** and the **FS** from the same instrument position is/are known as *intermediate sights* (IS).

- 2.500m and 0.500m are the respective readings observed on the staff at stations **A** and **B.**

- In all cases, the back-sight is observed on a known reduced level (RL), (Either already known or can be calculated).

Back-sight (BS) = 2.500m

Foresight (FS) = 0.500m

If RL at A = RL_A = 95.400m

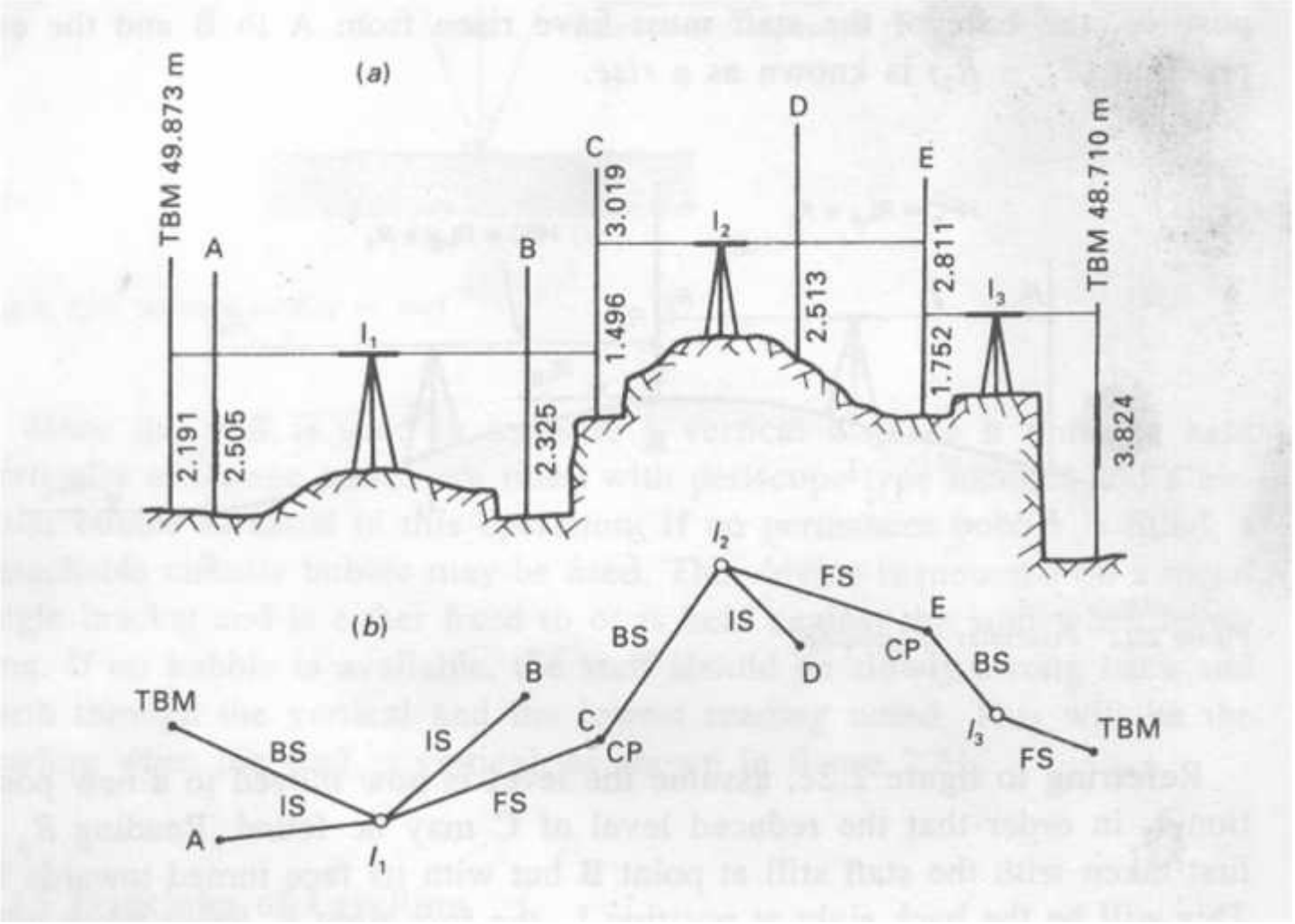
Then RL at B = RL_B = 95.400 + (2.500 - 0.500)

= 97.400m

In general RL is given by;

$RL_B = RL_A + (BS – FS)$

Example of levelling task



Field booking and reduction of levels

The Rise and Fall method

BS	IS	FS	Rise	Fall	Initial RL	Adjustment	Adjusted RL	Remarks
2.191					49.873	0	49.873	TBM 49.873
	2.505			0.314	49.559	+0.002	49.561	A
	2.325		0.180		49.739	+0.002	49.741	B
3.019		1.496	0.829		50.568	+0.002	50.570	C(CP)
	2.513		0.506		51.074	+0.004	51.078	D
1.752		2.811		0.298	50.776	+0.004	50.780	E(CP)
		3.824		2.072	48.704	+0.006	48.710	TBM 48.710
6.962		8.131	1.515	2.684	48.704			
8.131			2.684		49.873			
-1.169			-1.169		-1.169			

Arithmetic checks:

$$\Sigma (BS) - \Sigma (FS) = \Sigma(Rises) - \Sigma(Falls)$$

$$= \text{Last RL} - \text{First RL}$$

Adjustment

Adjustments are as follows:

- The difference between the calculated and known values of the RL (calculated RL – known RL) of the final Bench Mark (BM) is -0.006m.
- This is known as the misclosure.
- Since the misclosure is -0.006m, it implies that the total
- adjustment is +0.006m.
- This must be distributed to all the Calculated RL's
- Divide the adjustment by the total number of instrument positions or **BS.**

  =(+0.006/3) =+0.002m
- +0.002m is then added cumulatively to the calculated RL per every instrument station.

Height of Collimation method.

BS	IS	FS	HPC	Initial RL	Adjustment	Adjusted RL	Remarks
							TBM
2.191			52.064	49.873	0	49.873	49.873
	2.505			49.559	+0.002	49.561	A
	2.325			49.739	+0.002	49.741	B
3.019		1.496	53.587	50.568	+0.002	50.570	C(CP)
	2.513			51.074	+0.004	51.078	D
1.752		2.811	52.528	50.776	+0.004	50.780	E(CP)
							TBM
		3.824		48.704	+0.006	48.710	48.710

6.962	7.343	8.131	106.115	300.420
8.131				7.343 + 8.131 + 300.420 = 315.894
-1.169				(52.064 x 3) + (53.587 x 2) + 52.528 = 315.894
6.962				48.704
8.131				49.873
-1.169				-1.169

Arithmetic checks:

$$\Sigma (BS) - \Sigma (FS) = \text{Last RL} - \text{First RL}$$

$$\Sigma (IS) + \Sigma (FS) + \Sigma(\text{RL's except first}) = \Sigma(\text{each HPC x number of applications})$$

Calculation for HPC's and RL's

- Calculations to be performed for the height of collimation method are as follows:

- For instrument station I_1,

  HPC = 49.873 + 2.191 = 52.064m

  RL of A = 52.064 – 2.505 = 49.559m

  RL of B = 52.064 – 2.325 = 49.739m

  RL of C = 52.064 – 1.496 = 50.568m

- For instrument station I_2,

  HPC = 50.568 + 3.019 = 53.587m

The critical difference between Rise and Fall and HPC methods

- The rise and fall method is quicker to reduce where a lot of back sights and fore sights have been taken and very few intermediate sights taken

- The collimation method is quicker to reduce where a lot of intermediate sights have been taken since fewer calculations are required

Precision of levelling

- For a third order levelling, the allowable misclosure is given by:

$$\text{Allowable misclosure} = \pm 5\sqrt{n}\ mm$$

where *n* is the number of instrument positions

Or

$$\text{Allowable misclosure} = \pm 12\sqrt{k}\ mm$$

where ***k*** is the total distance levelled

Note:

The initial RL can only be adjusted if the allowable value is greater than the actual value.

Inverted staff

Sometimes the reduced levels of points above the height of instrument are required.

For example: the underside of a canopy; the levels of roof, caves, buildings etc.

- Generally, these points will be above the line of collimation.
- To obtain the reduced levels of such points, the staff is held upside down in an inverted position with its base on the elevated point.
- When booking an inverted staff reading it is entered in the levelling table with a minus sign.
- The calculation proceeds in the normal way, taking the minus sign into account.

Errors in levelling

- **Gross errors**
  - Wrong booking. (Eg. interchanging figures)
  - Omission or wrong entry. (eg. writing staff reading in the wrong column)
  - Spirit level not centred.

- **Constant errors**
  - Non-verticality of the staff
  - Collimation error in the instrument
  - Parallax

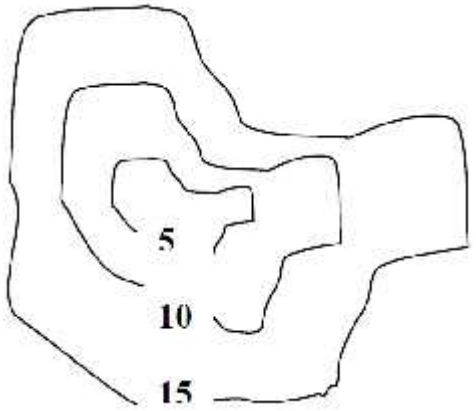
- **Random errors**
  - Effect of wind and temperature
  - Change point (*Firm spot must be chosen. For precise levelling, a base plate is used*)
  - Human deficiencies

Contouring

- Contour is by far the most extensively used method of showing relief features on a flat sheet of paper.

- This is because both the height and shape of the land surface can be specifically shown

- ***Contour lines*** are lines drawn on maps to join places of equal heights above or below sea level



Characteristics of contour lines

- Contour lines are continuous lines and cannot meet or cross any other contour line
- nor can any line split or join any other line, except in cases of cliff.
- Characteristically, contours are indicated by brown curved lines on maps.
- Every fifth contour line is thickened to facilitate easy reading.
- The interval between two contours, **i.e. *vertical interval (VI),*** is constant on the same map.
- Widely spaced contours indicate a gentle slope. Contours which are closer together indicate steep slopes and contours which are evenly spaced indicate uniform slope.

Contour Interval

- The difference in height between successive contours is known as the ***contour* or *vertical interval***

- This interval dictates the accuracy to which the ground is represented

- The value chosen for any application depends on:
  - The intended use of the plan
  - The scale of the plan (large scale = small contour interval)
  - The costs involved
  - The nature of the terrain.

Gradients

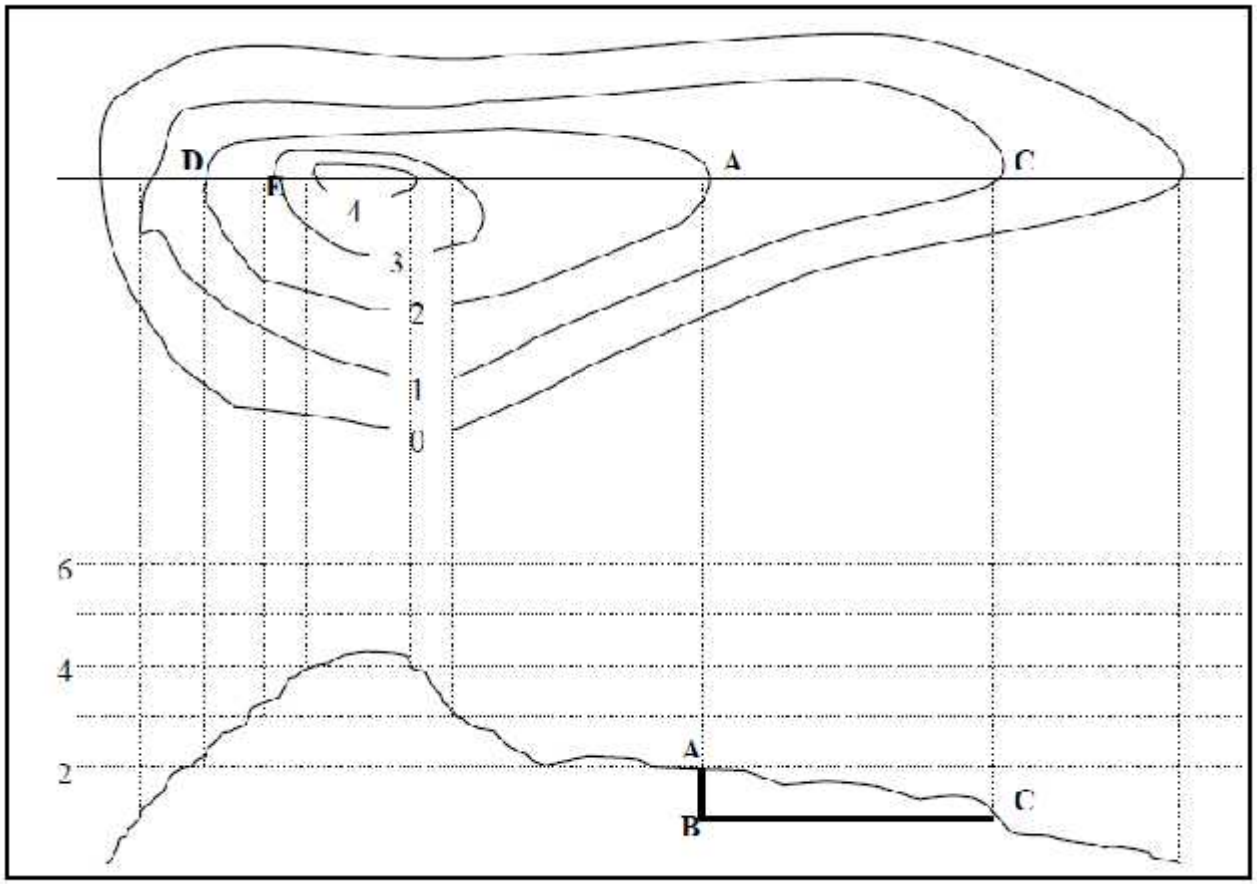
- The gradient of the ground between two points is given by:

$$\text{Gradient} = \frac{\text{Vertical Interval}}{\text{Horizontal Equivalent}}$$

- For example, the gradient of the ground between the points A and C is:

$$= \frac{AB}{BC} = \frac{\text{Vertical Interval}}{\text{Horizontal Equivalent}}$$

CROSS SECTION OF CONTOURS



Interpolating the contours

- This is done mathematically or graphically.

- With the mathematical method, the positions of the contours are interpolated mathematically from the reduced level by simple proportions.

- The height difference between each spot height is calculated and used with the horizontal distance between them to calculate the position on the line joining the spot heights at which the required contour is located